TheONE AI

Voice-First Artificial Intelligence Companion

Repository: github.com/Tarak098/TheONE-AI
Deployment: Render Web Service (FastAPI Full-Stack)
Document Version: 1.0 (Production Release)

1. EXECUTIVE SUMMARY

TheONE AI is a minimal, low-latency, voice-first companion inspired by ChatGPT's Advanced Voice Mode. Centered around an audio-reactive **Voice Orb**, the application replaces cluttered chatbot text feeds with real-time vocal conversation. It speaks through user-selected neural Indian voices and enforces a strict **one-sentence limit** (≤ 15 words) for razor-sharp comedic timing and context-aware roasts.

2. VOICE PIPELINE & ARCHITECTURE

```
[ User Microphone ]
  │  (Linear PCM 16-bit, 16 kHz Mono WAV)
  ▼
[ Gnani Prisma STT v3 ] ──► Recognized Text Transcript
  │
  ▼
[ Intelligence Orchestrator ]
  ├── Mode Prompt Resolution (Always Funny / Roast Companion)
  ├── Context-Aware Subject Alignment & Anti-Cliché Gate
  ├── Google Gemma 4 26B (w/ Gemini 2.5 Flash Failover)
  └── Strict Length Validator (≤ 1 Sentence, 3–10 Words)
  │
  ▼
[ Gnani Timbre TTS v2.5 ]
  ├── Always Funny: "Yashvi" (Female, Upbeat 1.04x speed)
  └── Roast Companion: "Zayan" (Male, Deadpan 0.98x w/ "..." pauses)
  │  (48 kHz WAV Container)
  ▼
[ Web Audio API Player ] ──► AnalyserNode ──► Reactive Voice Orb
```

3. PRODUCTION & DEPLOYMENT SPECIFICATION

Configured as a **unified full-stack ASGI service** on Render. FastAPI serves both the REST API endpoints and static frontend UI assets on the same origin, eliminating CORS issues and ensuring microphone access over HTTPS.

Platform	Render (Web Service / Blueprint `render.yaml`)
Runtime	Python 3.11 with Uvicorn ASGI
Build Command	`pip install -r requirements.txt`
Start Command	`uvicorn main:app --host 0.0.0.0 --port $PORT`
Health Check	`GET /health` → `{"status": "healthy"}`

4. CORE FEATURES & CAPABILITIES

- **Voice Orb (5 State Machine):** Visualizes `IDLE` (gentle breathing), `LISTENING` (mic volume reactive), `THINKING` (rotating aura), `SPEAKING` (acoustic ripples), and `ERROR` states.
- **Dual Personalities:** Always Funny (cheerful, sarcastic best friend) & Roast Companion (confident, savage roast master).
- **Independent Voice Selection:** Select between **Yashvi** (Female) and **Zayan** (Male) independently per mode with local storage persistence.
- **Strict Length Enforcement:** Enforces exactly 1 sentence, preferring 3–10 words (max 15 words) with zero paragraphs.
- **Context-Aware Humour:** Jokes/roasts directly target the user's specific subject; rejects disconnected stock clichés (e.g. random Wi-Fi or potato insults).
- **Safety & Sensitivity Protection:** Detects genuine grief or emergencies to respond with sincere empathy instead of roasting.
- **Drawer Controls:** Slide-over Conversation Transcript drawer, Settings drawer (voices & languages), and dark/light mode toggle.

5. TECHNICAL STACK

Backend	FastAPI 0.115+, Pydantic v2, Uvicorn
Frontend	Vanilla ES6+, HTML5 Canvas, Web Audio API, CSS3
Speech-to-Text	Gnani Prisma STT v3 REST API (16 kHz Mono)
Intelligence	Google Gemma 4 26B (Failover: Gemini 2.5 Flash)
Text-to-Speech	Gnani Timbre TTS v2.5 REST API (48 kHz WAV)
Audio Output	Yashvi (Female) & Zayan (Male) Neural Voices